

National University of Computer & Emerging Sciences

Quiz # 3(a)

Name & Roll Number: _____

Section: _____

Date: _____

Maximum Marks 10

1. A team has **8 machine learning engineers** and **6 data analysts**. A sub-team of **3 members** is selected.
 - (a) How many different sub-teams can be formed?
 - (b) How many sub-teams contain **at least one data analyst**?
2. A deck of 52 standard playing cards is shuffled. A single card is drawn at random. Define events: A : The card is a red card, B : The card is an ace card. Find $P(A \cup B)$ using the addition law of probability.
3. A selection algorithm picks **10 features: 5 numerical, 3 categorical, 2 Boolean**. If two are randomly chosen:
 - (a) Probability both are **categorical**.
 - (b) Probability first is **Boolean**, second is **numerical**.
4. A spam detection model filters emails from two sources: 70% emails from **corporate users**, 30% from **external users**, and **2.5%** of emails from corporate are incorrectly classified as spam, **5%** of emails from external are incorrectly classified as spam. If an email is flagged as spam, find the probability it came from an external source.

Question 1: Team Selection

Using the combination formula:

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

(a) **Total number of sub-teams**

Total team members:

$$8 + 6 = 14$$

Ways to choose a 3-member sub-team:

$$\binom{14}{3} = \frac{14!}{3!(14-3)!} = \frac{14 \times 13 \times 12}{3 \times 2 \times 1} = 364$$

(b) Sub-teams with at least one data analyst

Using the complement principle: - Sub-teams with only ML engineers:

$$\binom{8}{3} = \frac{8!}{3!(8-3)!} = \frac{8 \times 7 \times 6}{3 \times 2 \times 1} = 56$$

Thus, sub-teams with at least one data analyst:

$$\binom{14}{3} - \binom{8}{3} = 364 - 56 = 308$$

Question 2: Probability of $P(A \cup B)$

Using the addition rule of probability:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A) = \frac{26}{52} = 0.5$$

$$P(B) = \frac{4}{52} = \frac{1}{13}$$

$$P(A \cap B) = \frac{2}{52} = \frac{1}{26}$$

Now,

$$P(A \cup B) = \frac{26}{52} + \frac{4}{52} - \frac{2}{52} = \frac{28}{52} = \frac{7}{13}$$

Question 3: Feature Selection

Total features = 10 (5 numerical, 3 categorical, 2 Boolean)

(a) Probability both are categorical

$$P(\text{Categorical}, \text{Categorical}) = \frac{3}{10} \frac{2}{9} = \frac{1}{15}$$

(b) Probability first is Boolean, second is numerical

$$P(B, N) = P(\text{Boolean first}) \times P(\text{Numerical second})$$

$$= \frac{2}{10} \times \frac{5}{9} = \frac{10}{90} = \frac{1}{9}$$

Question 4: Spam Detection (Bayes' Theorem)

Using Bayes' Theorem:

$$P(E|S) = \frac{P(S|E)P(E)}{P(S)}$$

Given:

$$\begin{aligned} P(E) &= 0.3 & P(C) &= 0.7 \\ P(S|C) &= 0.025 & P(S|E) &= 0.05 \end{aligned}$$

Total probability of an email being flagged as spam:

$$\begin{aligned} P(S) &= P(S|C)P(C) + P(S|E)P(E) \\ &= (0.025 \times 0.7) + (0.05 \times 0.3) \\ &= 0.0175 + 0.015 = 0.0325 \end{aligned}$$

Now, compute:

$$\begin{aligned} P(E|S) &= \frac{P(S|E)P(E)}{P(S)} \\ &= \frac{0.05 \times 0.3}{0.0325} \\ &= \frac{0.015}{0.0325} \approx 0.4615 \end{aligned}$$

Final Answer: $P(E|S) \approx 46.15\%$

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Quiz # 3(b)

Name & Roll Number: _____

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Maximum Marks 10

1. A research team has **9 AI engineers** and **4 statisticians**. A group of **5** is selected.
 - (a) How many teams can be formed?
 - (b) Teams with **at least one statistician**?
2. A deck of 52 standard playing cards is shuffled. A single card is drawn at random. Define events: A : The card is a red card, B : The card is diamond card. Find $P(A \cup B)$ using the addition law of probability.
3. A feature selection algorithm picks **11 features: 5 numerical, 4 categorical, 2 Boolean**. If two are chosen:
 - (a) Probability both are **Boolean**.
 - (b) Probability first is **numerical**, second is **categorical**.
4. A spam detection model filters emails from two sources: 55% emails come from corporate, 45% from external and **3.5%** of emails from corporate are incorrectly classified as spam, **6.5%** of emails from external are incorrectly classified as spam. If an email is flagged as spam, find the probability it came from a corporate.

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Quiz # 3(c)

Name & Roll Number: _____

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Maximum Marks 10

1. A data science team consists of **7 machine learning engineers** and **5 data analysts**. A sub-team of **4 members** is to be selected at random.
 - (a) How many different sub-teams can be formed?
 - (b) How many sub-teams contain **at least one data analyst**?
2. A deck of 52 standard playing cards is shuffled. A single card is drawn at random. Define events: A : The card is a red card, B : The card is a face card (King, Queen, or Jack of any suit). Find $P(A \cup B)$ using the addition law of probability.
3. A feature selection algorithm selects **12 features**, consisting of **6 numerical**, **4 categorical**, and **2 Boolean features**. If two features are randomly selected **without replacement**, find:
 - (a) The probability that both selected features are **numerical**.
 - (b) The probability that the first feature is **numerical** and the second is **categorical**.
4. A spam detection model filters emails from two sources: 65% of emails come from **corporate users** (Source A). 35% of emails come from **external users** (Source B), and 2% of emails from Source A are incorrectly classified as spam, 5% of emails from Source B are incorrectly classified as spam. If an email is flagged as spam, what is the probability that it actually came from Source B?

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Quiz # 3(d)

Name & Roll Number: _____

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Maximum Marks 10

1. A data science team consists of **5 machine learning engineers** and **7 data analysts**. A sub-team of **3 members** is to be selected at random.
 - (a) How many different sub-teams can be formed?
 - (b) How many sub-teams contain **at least one data analyst**?
2. A deck of 52 standard playing cards is shuffled. A single card is drawn at random. Define events: A : The card is a black card, B : The card is a club card. Find $P(A \cup B)$ using the addition law of probability.
3. A feature selection algorithm selects **12 features**, consisting of **4 numerical, 6 categorical, and 2 Boolean features**. If two features are randomly selected **without replacement**, find:
 - (a) The probability that both selected features are **categorical**.
 - (b) The probability that the first feature is **categorical** and the second is **numerical**.
4. A spam detection model filters emails from two sources: 60% of emails come from **corporate users** (Source A). 40% of emails come from **external users** (Source B), and 2.3% of emails from Source A are incorrectly classified as spam, 5.1% of emails from Source B are incorrectly classified as spam. If an email is flagged as spam, what is the probability that it actually came from Source B?